SIEMENS Healthineers

Product Engineering Process (PEP), Directive
设计控制程序(PEP)

SSME
MI QR 4.4/01

55 21 906 - AND - 806 - 02

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Restricted

Siemens Shanghai Medical Equipment Ltd.

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Content

0. History / 修改历史

Nr.	Page	Version	Change description	CR No.
1	All	01	Initial creation for SSME MI QMS	N.A.
2	All	02	1. Add Chapter 5.9 Patents and Trademarks; 2. Update China RoHS requirements according to standard update in Chapter 2 and Chapter 5.13; 3. Add Chapter 5.16 Product Reliability Requirement; 4. Specify the process requirement in multiple chapters.	PCR11476

1. Purpose and scope / 目的和适用范围

Describes the SSME MI procedure for product development and provides information about necessary interactions to ensure an integrated, high quality, cost-effective, development process. This aligns with the Siemens Product Engineering Process (PEP) Model and, along with the referenced methods, guidelines, and templates, collectively represents the MI product development process.

描述 SSME MI 产品开发过程，并提供有关必要交互的信息，以确保集成、高质量、经济高效的开发过程。这与西门子产品设计控制程序 (PEP)模型一致，与引用的方法、指南和模板一起，共同代表了 MI 产品开发流程。

This procedure covers the PEP phases of Plan, Define, Realize, and Commercialize, but excludes any detailed description of Product Portfolio Management, which is primarily concerned with the planning of product families and the evolution of specific products based on overall business strategy, the competitive environment, and internal innovations.

此过程涵盖规划、定义、实现和商业化的 PEP 阶段，但不包括产品组合管理的任何详细描述，后者主要涉及产品系列的规划和基于整体业务战略、竞争环境和内部创新的特定产品的演变。

Depending on the particular project circumstances, these data can be adapted to requirements, as long as this does not run counter to the basic principle of the MI QR (tailoring is to be described in the QMP)

只要不违背质量保证程序基本原则，本规定可以根据项目的特殊情况进行修改以适合具体的需求(调整和裁剪 – 在质量管理计划中描述)。

This procedure applies all MI products.

此过程适用于 MI 所有产品。

2. Reference document / 参考文件

Document Number/ 文档编号	Document Name/文档名称
11106101-AND-02S	QR 0: Basic Quality Requirements/QR0：基本质量要求
ISO13485	Medical Devices – QMS /医疗器械–质量管理体系
IEC 62304 – Annex E	Medical Device Software – Software Life Cycle Processes/医疗器械软件–软件生命周期流程
4797663-AND-02S	QR 1: Identification of Products and General Labeling Requirements/QR1：产品标识和一般标签要求
5521906-AND-800	Quality Management Manual/质量管理手册
5521906-AND-801	Management Review Procedure/管理审查流程
5521906-AND-802	Management Responsibility/管理职责
5521922-AND-810	Customer Documentation WI/用户文档管理作业指导书
5521922-AND-813	Device History Content Requirements WI/产品追溯性内容要求作业指导书
5521922-AND-818	Product Phase-out Process at Product Lifecycle Management/产品生命周期管理中的产品逐步退出流程
5521906-AND-807	Product Risk Management Procedure/产品风险管理程序
5521922-AND-807	Software Development /软件开发
5521922-AND-809	Requirement Engineering Procedure Manual/需求工程程序手册
5521906-AND-808	Usability Engineering Process/可用性工程流程
5521906-AND-804	Regulatory Requirement/法规要求
5521922-AND-806	Product Verification and Validation/产品验证与确认
5521906-AND-808	Purchase Management Procedure for Production Material/生产材料采购管理程序
5521906-AND-805	Order Review Procedure/订单评审流程
5521906-AND-812	Process Validation and Verification Procedure/过程确认和验证流程
5521922-AND-805	Design Transfer WI/设计转移作业指导书
5521906-AND-821	Customer Complaint Handling Procedure/客户投诉处理程序
5521906-AND-809	Identification of Products and General Labelling Requirements/产品标识和标记的基本要求
5521906-AND-803	Change Control Procedure/修改控制流程
5521906-AND-826	Choosing and Applying the Statistical Technology Procedure/统计技术的选定和应用程序
11106101-AND-02S	QR0 - Basic Quality Requirements for Business Areas/Lines and Technology Excellence Units/ QR 0 －业务领域/生产线和技术卓越单元的基本质量要求

Measures for the Management of Restricted Use of Hazardous Substances in Electrical and Electronic Products 《电器电子产品有害物质限制使用管理办法》
GB/T 26572-2011 Requirements of Concentration Limits for certain restricted substances in Electrical and Electronic Products & GB/T 26572-2011 电子电气产品中限用物质的限量要求
GB 26572-2025 Requirements of Concentration Limits for certain restricted substances in Electrical and Electronic Products &GB 26572-2025 电子电器产品中限用物质的限量要求
GD 21-A8 10433621-A8-AND-02S-xx: Specific Requirements for China GD 21-A8 10433621-A8-AND-02S-xx: 对中国的特定要求
SJ/T 11364-2024 Labeling Requirements for Restricted Use of Hazardous Substances in Electrical and Electronic Product & SJ/T 11364-2024 电器电子产品有害物质限制使用标识要求

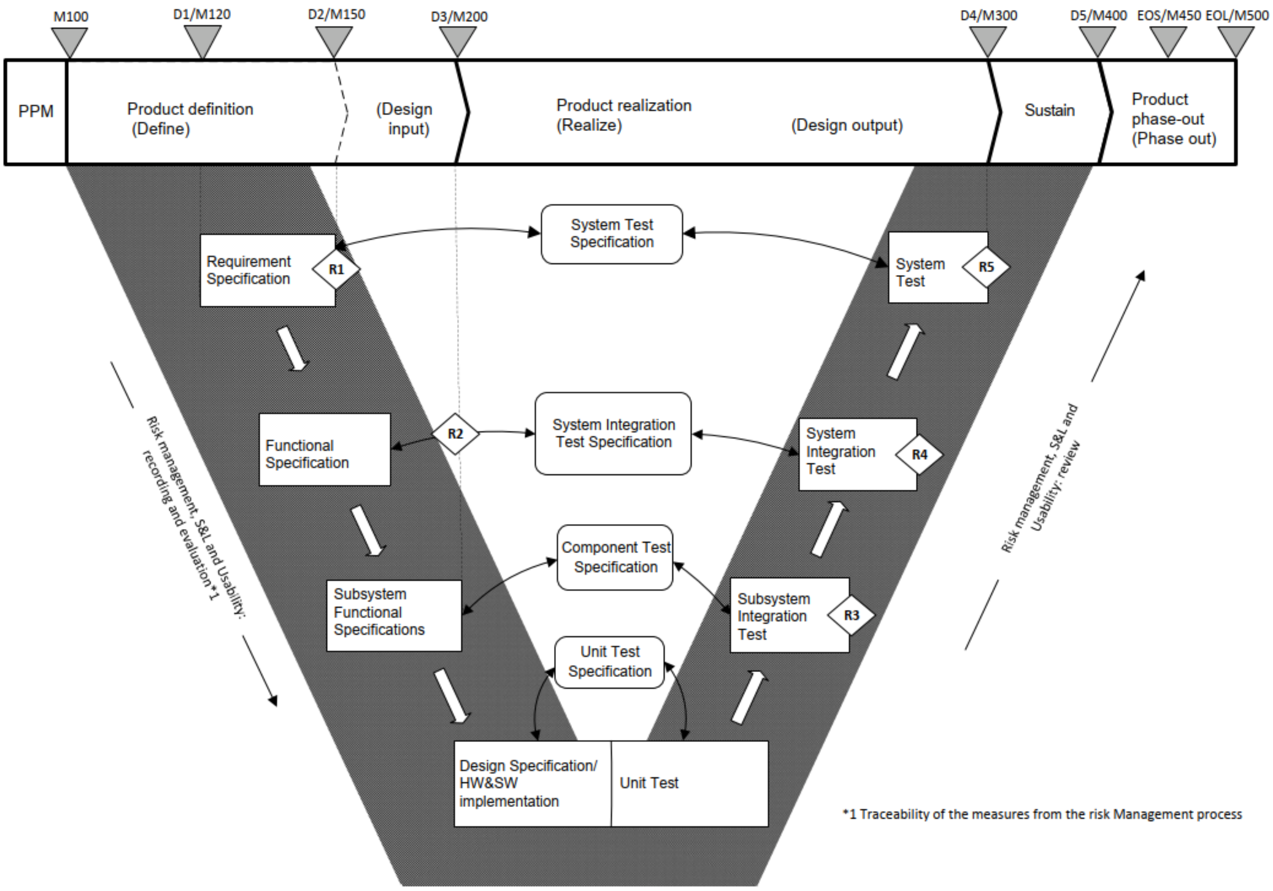
3. Abbreviations and Definitions / 缩略语与定义

RS	A document that defines the properties, characteristics, and capabilities of a system at the customer level, describing user needs and intended use. The requirements specification is an input for system test (design validation)	在客户层面定义系统属性、特性和功能的文档，描述用户需求和预期用途。需求规范是系统测试（设计确认）的输入。
FS	Functional Specification	功能规范
SubFS	Subsystem Functional Specifications	子系统功能规范
ST	System Test – test performed at the system level to ensure that the user-level requirements (as specified in the RS) are met.	系统测试–在系统级别执行的测试，以确保满足用户级别要求（如箱中指定）。
SIT	System Integration Test – test performed to ensure that the overall system meets its functional requirements.	系统集成测试–为确保整个系统满足其功能要求而执行的测试。
SSIT	Subsystem Integration Test – test performed to ensure that the subsystem meets its functional requirements	子系统集成测试–为确保子系统满足其功能要求而执行的测试。
PMP	Project Management Plan	项目管理计划
PPM	Product Portfolio Management	产品组合管理
QMP	Quality Management Plan	质量管理计划
QA	Quality Assurance	质量保证
CT	Computed Tomography	计算机断层扫描
MI	Molecular Imaging	分子影像
CS	Customer Service	客户服务
DS	Design Specifications	设计规范
SCM	Supply Chain Management	供应链管理
CRM	Customer Relationship Management	客户关系管理
EOS	End Of Product Support	产品支持结束
EOL	End Of Product Life cycle	产品生命周期结束
DHF	Design History File	设计历史记录文件
DHR	Device History Record	产品追溯性记录
DMR	Device Master Record	产品制造性文档
FMEA	Failure Modes and Effects Analysis	失效模式和效果分析
RoHS	Restriction of Hazardous Substances	关于限制在电子电气设备中使用某些有害成分的指令
EEP	Electrical and Electronic Product	电子电器产品
EFUP	Environmental Friendly Use Period	环保使用期限
PLM	Product Lifecycle Management	产品生命周期管理
PSG	Project Steering Group	项目领导小组
QR	Quality Regulation	质量保证程序
D1-D5	Main Decision	主要决定
R1-R5	Design Reviews	设计评审
SSME	Siemens Shanghai Medical Equipment Ltd.	上海西门子医疗器械有限公司
WI	Working Instruction	作业指导书

4. Procedure& Requirement / 过程和需求

4.1 V-model/Requirement Tracing / V 型模式/需求跟踪

This MI QR describes the product engineering process based on the V-model of product engineering.	本质量保证程序根据产品设计 V 字型模式说明产品设计过程。
The description of product engineering process in the following chapters shows the complete procedure from the systems perspective. Sub-systems / system components similarly follow the V-model.	本文以下章节所阐述的产品设计的完整步骤是针对系统的。对于子系统及系统部件，同样也要遵循这一 V 字形模式。



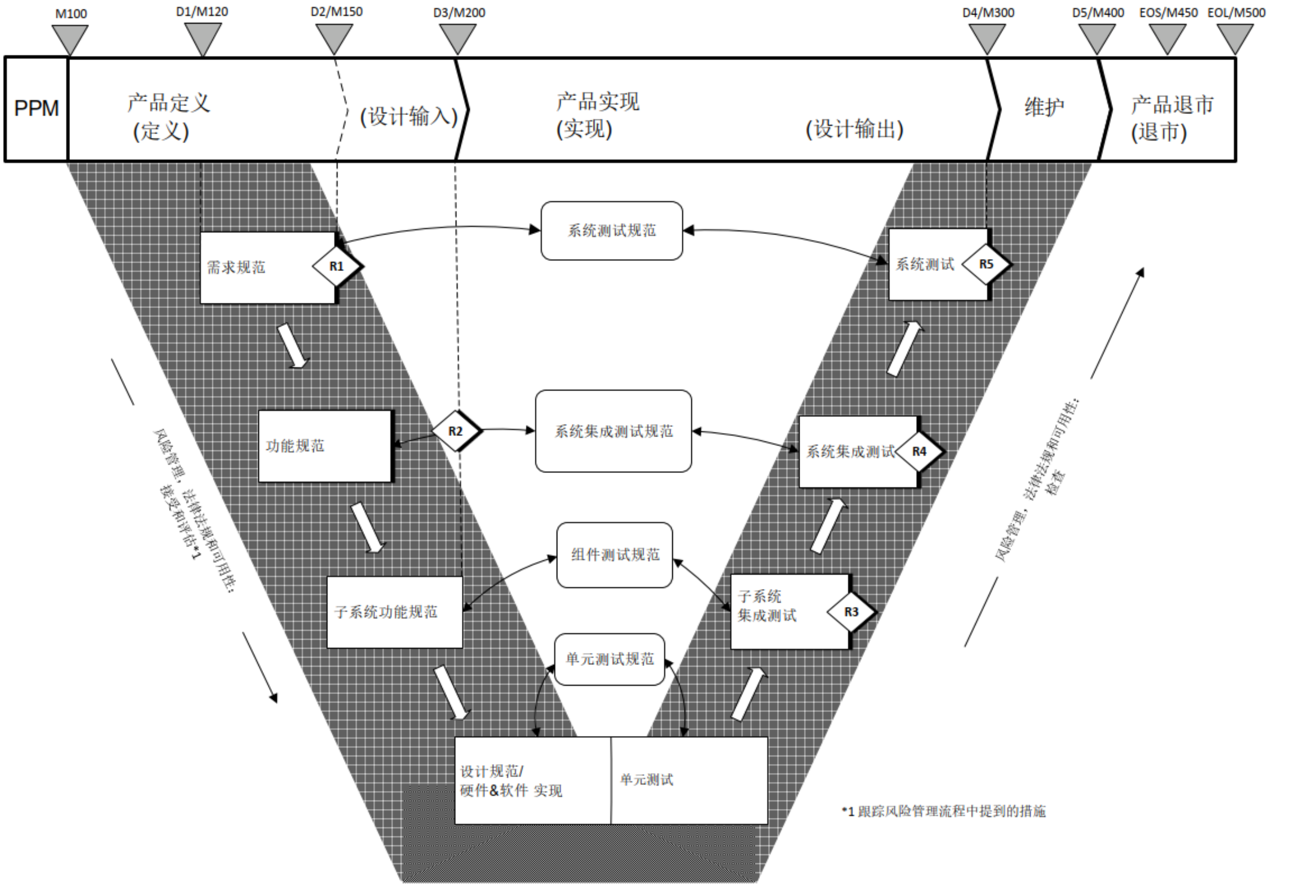


Fig.1/图 1: V-model/V 字型模式

Left branch of the V-model shows the product requirements are decomposed from Requirement Specification to Functional Specification, Subsystem Functional Specifications and then to Design Specification.	**V 字型模式左侧**表明产品需求从需求规范分解至功能规范，子系统功能规范，直到设计规范。
Right branch of the V-model presents the integration and test phases corresponding to the requirements.	**V 字型模式右侧**表明与需求相对应的集成和测试阶段。
Design and Development Planning Establish and maintain plans that describe or reference the design and development activities and define responsibility for implementation. The plans shall identify and describe the interfaces with different groups or activities that provide, or result in, input to the design and development process. The plans shall be reviewed, updated, and approved as design and development evolves.	**设计开发计划** 应建立和维护描述或参考设计和开发活动的计划，并定义实施责任。计划应识别和描述与不同小组或活动的接口，这些小组或活动为设计和开发过程提供或导致输入。随着设计和开发的发展，应对计划进行审查、更新和批准。
Design Input Establish and maintain procedures to ensure that the design requirements relating to a device are appropriate and address the intended use of the device, including the needs of the user and	**设计输入** 应建立和维护程序，以确保与设备相关的设计要求是适当的，并满足设备的预期用途，包括用户和患者的需求。这些程序应包括解决不完整、模棱两可或相互冲突的要求。设计输入要求应记录在案，并

patient. The procedures shall include a mechanism for addressing incomplete, ambiguous, or conflicting requirements. The design input requirements shall be documented and shall be reviewed and approved by a designated individual(s). The approval, including the date and signature of the individual(s)approving the requirements, shall be documented.	由指定人员审查和批准。批准，包括个人的日期和签名批准要求，应当记录在案。
Design Output Establish and maintain procedures for defining and documenting design output in terms that allow an adequate evaluation of conformance to design input requirements. Design output procedures shall contain or refer to acceptance criteria and shall ensure that those design outputs that are essential for the proper functioning of the device are identified. Design output shall be documented, reviewed, and approved before release. The approval, including the date and signature of the individual(s) approving the output, shall be documented.	**设计输出** 应建立和维护程序，以定义和记录设计输出，从而对设计输入要求的符合性进行充分评估。设计输出程序应包含或参考验收标准，并应确保识别出对设备正常运行至关重要的设计输出。设计输出应在发布前记录、审查和批准。应记录批准，包括批准输出的个人的日期和签名。
Design Verification Establish and maintain procedures for verifying the device design. Design verification shall confirm that the design output meets the design input requirements. The results of the design verification, including identification of the design, method(s), the date, and the individual(s) performing the verification, shall be documented in the DHF.	**设计验证** 应建立并维护验证装置设计的程序。设计验证应确认设计输出满足设计输入要求。设计验证的结果，包括设计识别、方法、日期和执行验证的个人，应记录在 DHF 中。
Design Transfer Establish and maintain procedures to ensure that the device design is correctly translated into production specifications.	**设计转移** 应建立和维护程序，以确保设备设计正确转化为生产规范。
Design Validation Establish and maintain procedures for validating the device design. Design validation shall be performed under defined operating conditions on initial production units, lots, or batches, or their equivalents. Design validation shall ensure that devices conform to defined user needs and intended uses and shall include test of production units under actual or simulated use conditions. Design validation shall include software validation and risk analysis, where appropriate. The results of the design validation, including identification of the design, method(s), the date, and the individual(s) performing the validation, shall be documented in the DHF.	**设计确认** 应建立并维护验证装置设计的程序。应在规定的操作条件下对初始生产单元、批次或批次或其等同物品进行设计确认。设计确认应确保设备符合规定的用户需求和预期用途，并应包括在实际或模拟使用条件下对生产单元进行测试。适当时，设计确认应包括软件验证和风险分析。设计确认的结果，包括设计识别、方法、日期和执行确认的个人，应记录在 DHF 中。

4.2 Structure of the Product Engineering Process / 产品设计过程结构

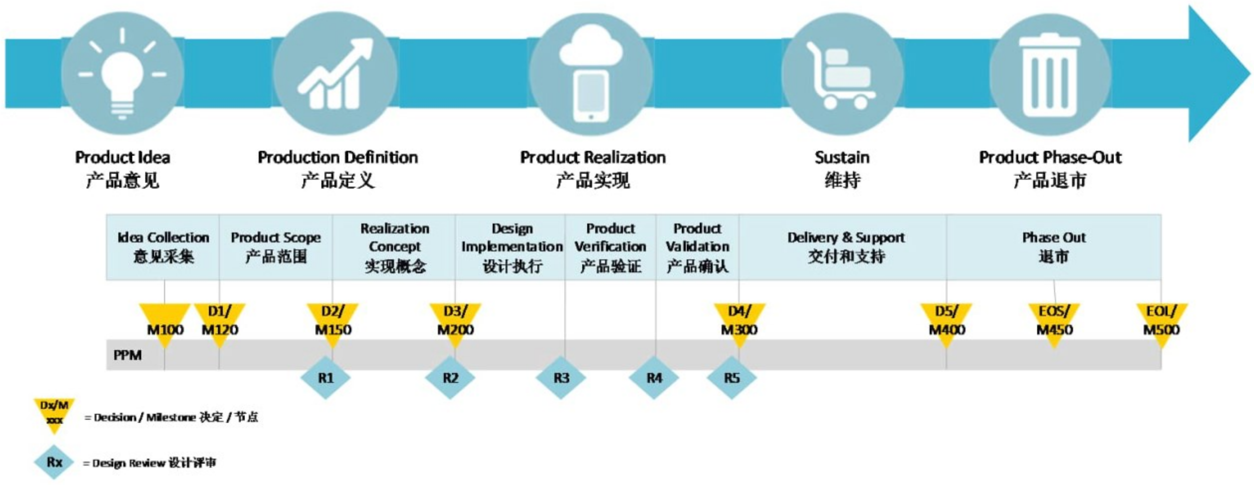


Fig. 2 / 图 2 产品设计过程结构

4.3 System development phases and their purpose / 系统开发阶段及其目的

4.3.1 Process Phase 1: Product Definition / 过程阶段 1：产品定义

Idea Collection (up to M120)	**意见采集（截止到M120）**
• Collection and evaluation of product ideas. • Analysis of product-specific market • Specification of business plan	• 产品想法收集和评估。 • 产品特定市场分析 • 商业计划的具体规范或详细说明
Product scope (M120 – M150)	**产品范围（M120- M150）**
Goal: • Definition of the product • Determination of market, customer, service, and SCM/ manufacturing requirements • Carry out product strategy and product profile planning • Where applicable, front-end trials, technology developments, clinical studies	目标： • 产品定义 • 确定市场、客户、服务以及供应链管理SCM/制造要求 • 执行产品策略和产品简介规划 • 在适用的情况下，进行前端试验、技术开发和临床研究

• Defining the **Product profile** by determining, defining, and prioritizing:
  ○ Market, customer, service, and SCM/manufacturing requirements, together with their systematic translation into hardware features (system functions) and software features (software functions).
  ○ Historical data for predecessor products,
  ○ Technical dependencies between hardware, eSW and software including all available platforms are evaluated.
	• **产品简介：**通过确定，定义和优先排序来定义。
	  ○ 对市场、客户、服务和供应链管理/制造需求，以及将其系统的转化为硬件指标（系统功能），软件指标（软件功能）。
	  ○ 此前产品的历史数据，
	  ○ 对硬件，嵌入式软件和软件间，包括所有可用平台的技术相关性进行评估。

• Defining the **Product concept**: by applying methods that take market and customer requirements into consideration for reaching the goals of:	• **产品概念：**通过采用考虑市场和客户需求的方法来定义，以达到：
	  ○ 实现战略业务目标，
	  ○ 缩短计划时间，更好的质量规划，

○ implementing strategic business objectives, ○ shorter planning times with a better planning quality, ○ a more efficient definition of product and system requirements. • Defining the **Product strategy** (recommended) by: ○ Do a market survey to analyze, amongst others, the potential and structure of the market (market volume, homogeneity, accessibility) and target groups (segmentation), ○ By analyzing and defining the key success factors of the new product, the potential and structure of the market (market volume, homogeneity, accessibility), the target groups (segmentation), its strengths, and weaknesses as well as restrictions of the organization. ○ Take into consideration existing strategic technology roadmaps as well as predefined strategies to create synergies.	○ 更有效的产品和系统需求定义。 • **产品策略**：（推荐通过以下定义） ○ 做市场调查，分析市场潜力和结构（市场容量，均匀性，可及性）和目标群体（市场划分）。 ○ 分析和定义新产品的关键成功因素，市场潜力和结构（市场容量、同质性、可接近性），以及目标群体（市场细分），优势和劣势，以及该组织的限制。 ○ 统筹考虑现有的战略技术路线图，以及预定义的策略，以创造协同效应。
Action:	**行动：**
1) Define the Project and Requirement Specification (Responsibility by Project Manager/Product Manager)	**1) 定义项目和需求规范（负责人：项目经理/产品经理）**
Definition of the product being delivered by the project is captured at a high level in the Project Management Plan and Product Manager specifies product project through one or more requirements specifications in detail. See 5521922-AND-809 for more information about requirements management. Requirements Specifications shall include Regulatory Requirement. (See 5521906-AND-804) Project managers coordinate with iteration planners to assure that specifications are available to the project when needed.	项目管理计划中对项目交付的产品进行高级别的定义，产品经理通过一个或多个需求规范对产品项目进行详细描述。有关需求管理的更多信息，请参阅5521922-AND-809。 需求规范需包含法规需求。（请参阅5521906-AND-804） 项目经理与迭代计划人员协调，以确保在需要时规范可用于项目。
2) Plan the Project (Responsibility by Project Manager)	**2) 规划项目（负责人：项目经理）**
Planning occurs during each of the project phases and the various plans are updated throughout the project lifecycle. The Project Management Plan, Quality Management Plan, and Risk Management Plan are the primary plans which drive the project. Note: Tailoring is permitted as described within the relevant process being used (pre-defined)	计划发生在每个项目阶段，并且在整个项目生命周期中更新各种计划。项目管理计划、质量管理计划和风险管理计划是推动项目的主要计划。 注意：允许按照正在使用的相关流程（预定义）中的说明进行定制。
a. Project Management Plan (Responsibility by Project Manager)	**a. 项目管理计划（负责人：项目经理）**
The Project Management Plan is the controlling document for managing the project. It also defines any tailoring of the process as agreed to by the project team.	项目管理计划是控制用于管理项目的文档它还定义了项目团队同意的任何流程裁剪。项目计划为充分的项目规划和控制提供了信息，包括

The Project Plan provides the information for adequate project planning and control, including milestones, the identification of team members, and the work products to be delivered. Note: The QMP and PMP may be combined in one document.	里程碑、团队成员的识别以及要交付的工作产品。 注意：QMP和PMP可以结合于一个文档中。
b. Software Naming (Responsibility by Project Manager)	**b. 软件命名（负责人：项目经理）**
As part of the planning process, software, and sometimes their associated platform projects, are given designations, as described in 5521922-AND-809. Designations are not typically assigned until shortly before R2. Documents being generated for R2 should use the appropriate designation if a designation is included.	作为规划过程的一部分，软件有时是其相关的平台项目所指定的，这在 5521922-AND-809 中有详细描述。名称通常直到 R2 之前不久才指定。 一旦指定，则为R2生成的文档应使用适当的名称。
c. Quality Management Plan (Responsibility by Quality Management Representative)	**c. 质量管理计划（负责人：质量管理负责人）**
The Quality Management Plan identifies the QA activities required to support the project. Quality controls specified in the QMP include participation in selected reviews, metrics gathering/reporting and performing R-reviews to ensure compliance with procedures.	质量管理计划确定了支持项目所需的质量保证活动。QMP中规定的质量控制包括参与选定的审查，指标收集/报告和执行R审查，以确保符合程序。
d. Risk Management Plan (Responsibility by Systems Engineering)	**d. 风险管理计划（负责人：系统工程师）**
The Risk Management Plan identifies how product risks are managed. It addresses how product risks will be identified, documented, and reported; and how mitigations for these risks will be identified, tracked, traced to appropriate documentation, and verified and validated. Product Risk Analysis begins early in the project and continues throughout the project lifecycle and sustaining. Product Risk Management Plan (See 5521906-AND-807)	风险管理计划确定如何管理产品风险。它解决了如何识别、记录和报告产品风险;以及如何识别、跟踪、追踪到适当的文档，并验证和确认这些风险的缓解措施。 产品风险分析开始于项目的早期，并在整个项目生命周期中持续进行维持。 产品风险管理计划(请参阅 5521906-AND-807)。
e. Other Project Level Planning Documents (As assigned by Project Manager)	**e. 其他项目级规划文件（项目经理指定）**
Reference documents: o Software Development (5521922-AND-807) o Usability Engineering Process (5521922-AND-808) o Regulatory Requirement (5521906-AND-804) o Product Verification and Validation (5521922-AND-806) o Purchase Management Procedure for	参考文件： o 软件开发（5521922-AND-807） o 可用性工程流程（5521922-AND-808） o 法规需求（5521906-AND-804） o 产品验证与确认（5521922-AND-806） o 生产材料采购管理程序（5521906-AND-808） o 需求工程程序手册（5521922-AND-

Production Material (5521906-AND-808) ○ Requirement Engineering Procedure Manual (5521922-AND-809)	809）
Realization concept (M150 – M200)	**实现概念（M150- M200）**
Goal: • Translation of product requirements into technical system requirements. • Definition of the system architecture • Identification of the product's potential hazards	目标： • 将产品需求转化为技术系统需求 • 定义系统架构 • 识别产品潜在的危害
Action:	**行动：**
3) Define functional specifications (Responsibility by Project Manager/PLM)	**3) 定义功能规范（负责人：项目经理/研发团队）**
The product backlog is assessed and updated on a regular basis to ensure that content is defined and prioritized according to the needs of the organization. These needs may address several facets (regulatory, market, environmental, technical, etc.) Iterations are relatively short-duration development activities. Content for an iteration is determined by retrieving items from the product backlog in priority order and according to the capabilities of the development team(s) involved. High-level product requirements from the product backlog are broken down and further defined, culminating in the creation of tasks. Tasks are assigned to individuals on the team for completion as part of a specific iteration. These tasks form the iteration backlog. Project managers work closely with iteration planners to ensure that content needed for the project is delivered when needed. Development provides functional specifications (FS, SubFS) as defined by the project's planning documents.	定期评估和更新产品待办事项列表，以确保根据组织的需要定义内容并确定其优先级。这些需求可能涉及几个方面（法规、市场、环境、技术等）。 迭代是持续时间相对较短的开发活动。迭代的内容是根据所涉及的开发团队的能力，按照优先级顺序从产品待办事项列表中检索项目来确定的。 来自产品待办事项列表的高级产品需求被分解并进一步定义，最终形成任务。任务被分配给团队中的个人，作为特定迭代的一部分来完成。这些任务构成了迭代的待办事项列表。 项目经理与迭代规划人员密切合作，以确保在需要时交付项目所需的内容。 开发提供由项目规划文档定义的功能规范（FS、SubFS）。

4.3.2 Process Phase 2: Product Realization / 过程阶段 2：产品实现

Design Implementation (M200 – R3)	**设计执行（M200- R3）**
Goal: • Development of the product (definition of design, implementation, verification) • Translation of requirements from subsystem requirement specification(s) into component requirement specifications, design descriptions, and designs	目标: • 产品开发（设计定义、实现、验证） • 从子系统需求规范中将需求转化为组件需求规范、设计描述和设计
Action:	**行动：**
1) Realization (Responsibility by PLM)	**1) 实现（负责人：研发团队）**
Development completes tasks assigned per the iteration backlog. Design Specifications (DS) are created as needed to show how the hardware and/or software system will be structured to satisfy	开发完成每个迭代待办事项安排分配的任务。设计规范(DS)是根据需要创建的，以显示硬件和/或软件系统将如何构建，以满足分配给该系统的需求。DS 包括对实现所需的结构、接口和数据的描

the requirements allocated to that system. The DS includes a description of the structure, interfaces, and data necessary for implementation.	述。
2) Test(Responsibility by Hardware Engineer/Software Engineer)	**2) 测试（负责人：硬件/软件工程师）**
Unit and Subsystem Integration test is performed within iterations. Automated test is utilized when possible. (See 5521922-AND-807)	单元和子系统集成测试在迭代中执行。尽可能使用自动测试。（请参阅 5521922-AND-807）
a. Perform Unit Test (Responsibility by Hardware Engineer/Software Engineer)	**a. 执行单元测试（负责人：硬件/软件工程师）**
Unit test (as applicable) is done at the lowest level of implementation, e.g., the unit or component, to verify that the unit meets the requirements allocated to the unit / component. o Unit Test Specification (See 5521922-AND-807)	单元测试（如适用）是在最低的实现级别（例如单元或组件）完成的，以验证单元是否满足分配给单元/组件的要求。 o 单元测试规范（请参阅 5521922-AND-807）
b. Perform Subsystem Integration Test (Responsibility by Hardware Engineer/Software Engineer)	**b. 执行子系统集成测试（负责人：硬件/软件工程师）**
Referred to as SSIT, this activity consists of verification that the subsystems meet the functional requirements that have been allocated to them from higher level specification(s) A subsystem consists of one or more functional units. o Subsystem Integration Testing Specification (See 5521922-AND-806)	此活动称为 SSIT，包括验证子系统是否满足从更高级别规范分配给它们的功能要求。子系统由一个或多个功能单元组成。 o 子系统集成测试规范（请参阅 5521922-AND-806）
Product verification (R3– R4)	**产品验证（R3-R4）**
From PreR1 to R3, project activities work in parallel to and synchronized with the continuous development stream. From R3 onwards, the project activities become separate, independent activities to verify, validate and release the product for production. Resources on the development stream are utilized as needed in support of these activities.	从 R1 之前到 R3，项目活动与持续开发流并行并同步。从 R3 开始，项目活动成为单独的、独立的活动，以验证、确认和发布产品以供生产。 根据需要利用开发流上的资源来支持这些活动。
Goal: Proof of realization of technical requirements. The system and its documentation including accompanying documents are ready for product validation.	目标：技术要求实现的证明。系统及其文档，包括随附文件已准备好进行产品验证。
Action:	**行动：**
3) Perform System Integration Test (Verification) (Responsibility by Hardware Engineer/Software Engineer/Systems Engineering)	**3) 执行系统集成测试（验证）（负责人：硬件/软件/系统工程师）**
Referred to as SIT, this activity consists of verifying that the system meets the functional requirements that have been defined for it. System Integration Test Specification (See 5521922-AND-806) Perform final verification (system verification): Verification test reports (depending on test	此活动称为 SIT，包括验证系统是否满足为其定义的功能要求。 系统集成测试规范（请参阅 5521922-AND-806） 执行最终验证（系统验证）：验证测试报告（根据测试概念：子系统/系统级别的报告；包括安全测试和可靠性测试的报告）。

concept: reports for subsystem/system level; including reports for safety tests and reliability tests)	
4) Perform Usability Test (Responsibility by Usability Lead/Systems Engineering)	**4) 执行可用性测试（负责人：可用性负责人/系统工程师）**
See 5521922-AND-808 for details about usability test.	有关可用性测试的详细信息，请参阅 5521922-AND-808
5) Perform Design Transfer (Responsibility by Hardware Engineer/Software Engineer/Systems Engineering)	**5) 执行设计转移（负责人：硬件/软件/系统工程师）**
This activity ensures that the product is transferred to manufacturing for production. PLM provides the key values that SCM is required to use as part of manufacturing test procedures. • Design Transfer WI (5521922-AND-805）	此活动可确保将产品转移到制造部门进行生产。PLM 提供 SCM 在制造测试过程中需要使用的关键参数。 • 设计转移作业指导书（5521922-AND-805）
Product validation (R4 – M300)	**产品确认（R4-M300）**
Goal: • Proof of realization of product (/ technical) requirements. • Final evaluation of usability and the specified intended use/intended purpose of product. • The system and its documentation including accompanying documents are ready for product validation.	目标： • 产品（/技术）要求实现的证明。 • 对产品可用性的最终评估以及对产品指定的预期用途/预期目的的评估。 • 系统及其文档，包括随附文件，已准备好进行产品验证。
Action:	**行动：**
6) Perform System Test (Validation) (System Test)	**6) 执行系统测试（确认）（系统测试）**
Referred to as ST, this activity focuses on validating the integrated system against the requirements of the Requirements Specification. Test is performed from a customer perspective to validate the system for its intended use. System Test Specification (See 5521922-AND-806) • Design is validated (see section 4.1), including a final review of the effectiveness of the required measures derived from the risk analysis：system validation test report, risk management report, formative and summative user interface evaluation report. • Reliability measures defined in reliability engineering plan are implemented and verified. Reliability engineering report is written.	此活动称为 ST，重点是根据需求规范的要求确认集成系统。测试是从客户的角度执行的，以确认系统的预期用途。 系统测试规范（请参阅 5521922-AND-806） • 对设计进行确认（见第4.1部分），包括风险分析所要求的防范措施的有效性最终评审：系统确认测试报告，风险管理报告，形成性与总结性用户界面评估报告。 • 可靠性工程计划中定义的可靠性措施得到实施和验证。编写可靠性工程报告。
7) Perform Process Validation (Responsibility by SCM)	**7) 执行过程确认（负责人：SCM）**
Series production samples are created on the basis of the approved production documents (including process validation) • Process Validation and Verification Procedure	基于已批准的生产文档（包括过程确认），创建批量生产样品。 • 过程确认和验证程序（5521906-AND-812）

4.3.3 Process Phase 3: Sustain and Product Phase out / 过程阶段 3：维护和产品退市

Delivery & Product support (M300 – M400)	**交付和产品支持（M300–M400）**
Goal: • Delivery of error free products • Product support • Preparation of the phase out	目标： • 无缺陷产品交付 • 产品支持 • 产品退市准备
Action:	**行动：**
1) Support SCM (Responsibility by Hardware Engineer/Software Engineer/Systems Engineering)	**1) 支持SCM（负责人：硬件/软件/系统工程师）**
In this phase, PLM supports the SCM process with technical advice and data which are collected, monitored, and prepared during commercialization.	在此阶段，PLM通过技术建议和数据来支持SCM流程，这些建议和数据在商业化过程中收集、监控和准备。
a. Provide and receive SCM information (Responsibility by Hardware Engineer/Software Engineer/Systems Engineering)	**a. 提供和接收SCM信息（负责人：硬件/软件/系统工程师）**
Changes are coordinated through Change Control Board (See 5521906-AND-803) Small changes require only minimal development activities along with appropriate verification and validation. Large changes may require a full development project or could be incorporated into an existing project.	变更通过修改控制委员会（请参阅 5521906-AND-803）进行协调。小的修改只需要最少的开发活动以及适当的验证和确认。重大修改可能需要完整的开发项目，或者可以合并到现有项目中。
2) Support CRM (Responsibility by Hardware Engineer/Software Engineer/Systems Engineering)	**2) 支持客户关系管理（负责人：硬件/软件/系统工程师）**
In this phase, PLM supports the CRM care and sell processes with all relevant product information and data which are collected, monitored, and prepared during the commercialization phase.	在此阶段，PLM通过所有相关的产品信息和数据来支持CRM关怀和销售流程，这些信息和数据在商业化阶段被收集、监控和准备。
a. Investigate Complaints (Responsibility by Hardware Engineer/Software Engineer/Systems Engineering)	**a. 调查投诉（负责人：硬件/软件/系统工程师）**
Complaints are investigated and, where necessary, change requests are issued. If applicable, a root cause analysis is done. ‘Investigate Complaints’ is performed until End of Support (EOS), for hazards until the End of Life (EOL) of the product. Customer Complaint Handling Procedure (See 5521906-AND-821)	对投诉进行调查，并在必要时发出更改请求。如果适用，将进行根本原因分析。"调查投诉"在支持终止（EOS）之前执行，对危险（源）执行到产品生命周期结束（EOL）。 客户投诉处理程序（请参阅 5521906-AND-821）
Discontinuation of production (M400 – M450)	**生产停止（M400–M450）**
Goal: The phase out of a product is initiated, end of support is reached.	目标：启动产品退市，停止产品支持。
Product phase-out (M450 – M500)	**产品退市（M450 – M500）**
Goal: • Phase out of a product is initiated • End of support is reached	目标: • 启动产品退市 • 停止产品支持
• Discontinuation of production: The phase out of a product is initiated, end of support is	• 生产停止：启动产品退市，停止产品支持。

reached. • Phase-out: The phase out of a product is performed; end of product lifecycle is reached. See 5521922-AND-818 for details about Product Phase-out Process at Product Lifecycle Management.	• 退市：产品退市实行，产品生命周期结束。 有关产品生命周期管理中的产品逐步退出流程，请参阅 5521922-AND-818

4.4 Review items and milestones in the PLM / 评审和里程碑

The milestones in the PLM are defined as follows:

- M100: Decision to start the process section product definition
- M120(D1): Decision to initiate the product definition
- M150(D2): Decision to commit the realization concept phase
- M200(D3): Decision to commit the design and development
- M300(D4): Delivery release (series)
- M400(D5): Decision about end of delivery
- M450 (EOS): End of the guaranteed spare parts supply period (end of product support)
- M500(EOL): End of product lifecycle

定义如下项目里程碑：

- M100：决定启动产品定义流程
- M120(D1)：决定开始产品定义
- M150(D2)： 决定启动产品实现方案阶段
- M200(D3)： 决定确认设计和开发阶段
- M300(D4)：产品发运的决定
- M400(D5)：停止发运的决定

  M450 (EOS)：决定结束配件供应的日期（结束产品的支持）
- M500(EOL):：决定终止产品生命周期

Design Review	**设计评审**
Establish and maintain procedures to ensure that formal documented reviews of the design results are planned and conducted at appropriate stages of the device's design development. The procedures shall ensure that participants at each design review include representatives of all functions concerned with the design stage being reviewed and an individual(s) who does not have direct responsibility for the design stage being reviewed, as well as any specialists needed. The results of a design review, including identification of the design, the date, and the individual(s) performing the review, shall be documented in the design history file (the DHF) Detailed description of Project Design Review is stated in 5.2 Design and Document Reviews.	应建立并维护程序，以确保在设备设计开发的适当阶段计划并进行设计结果的正式文件审查。程序应确保每次设计评审的参与者包括与所审查的设计阶段相关的所有职能部门的代表、对所审查设计阶段没有直接责任的个人以及所需的任何专家。设计评审的结果，包括设计标识、日期和进行审查的人员，应记录在设计历史文档（DHF）中。 项目设计评审的描述请参阅 5.2 设计和文档审查。
R Reviews are project quality review points(See Figure 1), the R-Review Checklist (Appendix 1) is the definitive list of required deliverables for each	R 审查是项目质量审查点（见图 1），R-Review 清单（附录 1）是每个 R-Review 所需可交付结果的最终列表和审核的职责人员。下表将 R-Review 映射到相关的

R Review and responsible pesrson. The table below maps the R-Reviews to related project activities.	项目活动。

R-Reviews	Definition	定义
R1	To confirm that the user requirements are adequately defined, and the project management structure is in place. The project can proceed to the R2 milestone.	确认用户需求已充分定义，项目管理结构已到位。该项目可以进入 R2 里程碑。
R2	To confirm that the system is adequately defined and the tasks to complete the project are understood. The project can proceed to the R3 milestone.	确认系统已充分定义并了解完成项目的任务。该项目可以进入 R3 里程碑。
R3	To verify that implementation and Subsystem Integration Test (SSIT) have been completed	验证实施和子系统集成测试（SSIT）是否已完成。
R4	To verify that SIT has been completed	验证 SIT 是否已完成
R5	To confirm that the product is ready for release.	确认产品已准备好发布。

5. General requirements / 通用要求

5.1 Tailoring / 裁减

5.1.1 Project Specific Tailoring / 项目特定裁减

Depending on individual project circumstances, the requirements and stipulations in this directive may be adapted in line with particular needs provided that there is appropriate justification (tailoring). However, any such changes must not compromise the basic principles of the PEP. In the case of further developments, for example, the formal M120(D1) and M150(D2) decisions may be omitted, providing that all the required design documents are already available. Further project-specific milestones may be necessary in order to ensure that the project documentation is produced in a sensible order. The reasons for the tailoring must be documented in the quality management plan.	根据具体项目情况，在由适当理由的情况下，可以依照特定需求对指令的需求和规定进行调整（裁减）。但任意此类变更均不得与 PEP 的基本原则相违背。进一步开发情况下，如所有要求的设计文档均已可用，则可省略正式 M120（D1）和 M150（D2）决定。可能需要针对具体项目设立节点，以确保项目文档按合理顺序生成。裁减的理由必须记录在质量管理计划中。

5.1.2 Software Specific Tailoring / 软件特定裁减

Additional instructions apply in the case of software development.	适用于软件开发的其它说明。

5.1.3 Standard Tailoring for Successor Versions of Platform Release/ 平台发布后续版本的标准裁减

The first version of a platform is developed with all milestones (M120, R1, M150, R2, M200). Standard Tailoring: For successor projects R1/R2 and M120/M150/M200 can be combined to one R and one M milestone: R2/M200. Required milestone inputs are the sum of R1 and R2.	平台的第一个版本开发经历所有主评审（M120，R1，M150，R2，M200）。 标准裁减：对于后续项目 R1 / R2 和 M120 / M150 / M200 可以组合为一个 R 和一个 M 主评审：R2 / M200。所需的评审输入是 R1 和 R2 的总和。

5.1.4 Standard Tailoring for Agile Approaches during Product Development / 产品开发期间敏捷方法的标准裁减

Permissions: a. R1/R2 and M120/M150/M200 are combined to one R and one M milestone. Requirement specifications like Market Requirements Specification or Engineering Requirements Specification are not expected as milestone input but planned as a committed product backlog. They will be created according to backlog planning. Details for further milestone inputs are defined in the Quality Management Plan	权限： a. R1/R2 和 M120/M150/M200 合并为一个 R 和一个 M 主评审。需求规范（如市场需求规范或工程需求规范）不是里程碑输入预期的，而是规划承诺的产品待办事项。它们将根据待办工作计划创建。质量管理计划中定义了更多评审输入的详细信息。
a. Further process results are generated step by step according to the prioritized product backlog. Meanwhile the order and the dependencies of process results must be considered. The dependencies are visible on SOP level.	a. 进一步的过程结果将根据优先排序的产品待办事项逐步生成。同时，必须考虑过程结果的顺序和依赖关系。依赖关系在 SOP（标准操作程序）层面上是可见的。

5.2 Design and Document Reviews / 设计和文档审查

Design and Document Reviews are held to ensure the adequacy, technical feasibility, and completeness of the specifications, design, or other deliverable being reviewed. Note: When releasing related documents, it is recommended that project teams release documents in the order of dependence. Generally, this means that upper-level requirements are defined and released prior to lower-level requirements, and that requirements are defined and released prior to their associated test specifications.	设计和文档评审是为了确保被评审的规格、设计或其他可交付成果的充分性、技术可行性和完整性。 注意：发布相关文档时，建议项目团队按依赖关系的顺序发布文档。通常，这意味着上层需求是在下层需求之前定义和发布的，而需求是在他们指定的测试规范之前定义和发布的。

5.3 Change Management / 变更管理

Design Change Establish and maintain procedures for the identification, documentation, validation or where appropriate verification, review, and approval of design changes before their implementation.	**设计变更** 应在实施设计变更之前，建立并维护设计变更的识别、文件编制、验证或适当的验证、审查和批准程序。
Product Change Management within the scope of development (e.g., not the Engineering Change Order process) is governed by the Product Change Control process and facilitated by Change Control Board. These include, but are not limited to, changes to design documents and requirements, and to managing defects. • Change Control Procedure (See 5521906-AND-803)	开发范围内的产品变更管理（例如，不是工程变更单流程）由修改控制程序管理，并由修改控制委员会提供便利。 这些包括但不限于对设计文档和要求的更改，以及对缺陷的管理。 • 修改控制程序（请参阅 5521906-AND-803 ）

5.4 Document Management / 文档管理

Documents which are required to be maintained as part of the development process (for example, Plans, Specifications, Test Results, etc.) are governed by quality processes.	作为开发过程的一部分需要维护的文档（例如，计划、规范、测试结果等）受质量流程的约束。

5.5 Traceability Matrix / 可追溯性矩阵

Traceability matrices are used during the development process to confirm that all requirements have been addressed both in design and test. Note: Requirements that are mitigating hazards and allocated to software, as well as software that has been classified as Class C per IEC 62304, must be traceable into the code itself. The	在开发过程中使用可追溯性矩阵来确认在设计和测试中都已满足所有要求。 注意：减轻危险（源）并分配给软件的要求，以及根据 IEC62304 被归类为 C 类的软件，必须可追溯到代码本身。代码的可追溯性可以通过跟踪矩阵或其他替代手段建立。缓解危险（源）且未分配给软

traceability to code may be established with a trace matrix or by other alternate means. Requirements that mitigate hazards and are not allocated to software must be traceable to the lowest level of implementation of the mitigation within the FS, SubFS, or DS.	件的要求必须可追溯到 FS，SubFS 或 DS 中缓解措施的最低实施级别。

5.6 Design History File / 设计历史文档

A DHF shall be established and maintained for each type of device. The DHF shall contain or reference the records necessary to demonstrate that the design was developed in accordance with the approved design plan and the requirements of this part.	应为每种类型的装置建立并维护设计历史文档。设计历史文档应包含或参考必要的记录，以证明设计是根据批准的设计计划和本部分的要求制定的。

5.7 Relationship of Projects to the Development Stream / 项目与开发流的关系

Projects are scheduled by the business to deliver products to support the business strategy. Since the development stream is continuous, the projects are synchronized with the development stream via the product backlog and the project schedule. Multiple projects can run in parallel, as needed, to support this strategy.	项目由业务部门安排，以交付支持业务战略的产品。由于开发流是连续的，因此项目通过产品待办事项列表和项目进度与开发流同步。根据需要，多个项目可以并行运行，以支持此策略。

5.8 Standards & Laws / 标准和法律

Compliance with laws and standard (See 5521906-AND-804 Regulatory requirement) must be ensured over the entire life cycle of the product. Adjustments to the product must be assessed and measures implemented if standards or laws have been changed.	在产品的整个生命周期必须符合法律和标准（请参阅 5521906-AND-804 法规需求）的要求。如果法律法规有更改，需要对产品做相应的调整评估，并实行应对措施。

5.9 Patents and trademarks/ 专利和商标

Patent and Trademark rights must be considered: • IP clearing need to be conducted for new products and new features. • New trademark rights respectively should be applied. • Third-party trademark rights must be considered and evaluated.	需考虑专利和商标权： • 对新产品及新功能需要开展专利澄清。 • 新的商标权可分别申请。 • 需考虑并评估第三方商标权。

5.10 Procurement management / 采购管理

The procedure outlined in (See 5521906-AND-808 Purchase Management Procedure for Production Material) for e.g., supplier evaluation, specification, verification, and documentation must be followed for internal and external	在项目实施过程中，系统和软件组件、外部工作或服务的内部和外部采购必须遵循（请参阅 5521906-AND-808 生产材料采购管理程序）中概述的程序，如供应商评价、规范、验证和文件。

purchases of system and software components, external work, or services during the execution of the project.	

5.11 Preventive action / 预防措施

Information from predecessor products must be taken into consideration when developing new products or refining existing products in order to systematically reduce the possibility of faults or errors. The results of appropriate investigations must be documented (error prevention report). Inputs are e.g. change request, complaints or system, design and process FMEA.	在开发新产品或者改进现有产品时，必须考虑之前产品的信息，以保证系统性地减少缺陷和错误，相关的调查结果必须记录在文档中（错误预防报告）。 例如，更改，抱怨，及系统、设计、流程的FMEA。

5.12 Product-Related Environmental Protection/项目相关环境保护

The requirements regarding project-related environmental protection are aimed at the efficient implementation of product-related decisions and to mitigate concerns. Specific, measurable, and achievable environmental criteria shall be defined, that support the creation of environmentally sustainable products. These criteria are integrated into the standard development process.	关于项目相关环境保护的要求旨在高效执行与产品相关的决策，并减轻担忧。 应定义具体、可测量且可实现的环境标准，以支持创建环境可持续的产品。这些标准被整合到标准开发流程中。

5.13 China RoHS Requirement / 中国 RoHS 要求

5.13.1 Product Scope 产品范围	
Electrical and electronic products in the scope of China RoHS requirements are as follows: •Devices and accessories that operate on electric current or electromagnetic fields or for the purpose of generating, transmitting and measuring electric current and electromagnetic fields, with rated operating voltages not exceeding 1500 V DC and 1000 V AC. •Parts, components and sub-assemblies for use in electrical and electronic products within the scope of application of the Scheme. Electrical and electronic products not in scope of China RoHS requirements are as follows: •Products manufactured in China for export. China RoHS requirements do not apply to Hong Kong, Macao and Taiwan. •Products temporarily imported into China or brought in for repair, but not for sale. •Products used for scientific research/development, testing purposes. •Products used for exhibitions, fairs, etc., but not for sale. •Products not sold into China	中国 RoHS 范围的电子电器产品包括： -依靠电流或磁场工作或者以产生、传输和测量电流和电磁场为目的，额定工作电压直流电不超过 1500V、交流电不超过 1000V 的设备及配套产品。 -包括适用范围内的电子电器产品中所使用的零部件及子部件。 不在中国 RoHS 范围内的 - 在中国制造并从中国出口的设备，中国 RoHS 要求不适用于香港、澳门和台湾地区。 - 为了维修而非销售的目的，临时进口至中国或引入中国的产品。 - 用于科学研究、测试的产品。 - 用于展览、展会等场合但不销售的产品。 - 不在中国销售的产品。 - 电池供电的温度/相对湿度或其他运输条件测量设备，该设备包含在包装/运输材料中，但不属于安装在客户现场的设备组成部分。

•Battery-powered temperature/relative humidity or other shipping condition measurement devices that are included within packaging/shipping materials but that are NOT a part of the device installed at the customer site

5.13.2 Products classification and requirements 产品分类及要求

-Classification of electrical and electronic products and restricted use requirements for hazardous substances， see Table [1]

Table 1

Category	Product Classification	China RoHS Requirements
I	Products included in the “Catalog of management of restricted use of hazardous substances in electrical and electronic products“	Concentration limit requirements; Labeling requirements
II	Products not included in the “Catalog of management of restricted use of hazardous substances in electrical and electronic products“	Labeling requirements
Medical device and In-Vitro Diagnostic (IVD) are classified as Class II products		

-电器电子产品分类及有害物质限制使用要求,见表 1

表 1

电器电子产品类别	电器电子产品分类	中国RoHS要求
I	纳入《电器电子产品有害物质限制使用达标管理目录》的产品	含量限制要求;标识要求
II	未纳入《电器电子产品有害物质限制使用达标管理目录》的产品	标识要求
医疗器械和体外诊断（IVD）产品被归类为II类产品		

-Requirements of China RoHS restricted substances concentration limits see GB/T 26572 and GB 26572 are as follows: the content of lead (Pb), mercury(Hg), hexavalent chromium (Cr6+), polybrominated biphenyls (PBB), polybrominated diphenyl ethers (PBDE) ,Dibutyl phthalate(DBP) , Diisobutyl phthalate (DIBP) , Butyl benzyl phthalate (BBP) , Bis(2-ethylhexyl) phthalate (DEHP)shall be less than 0.1% and the content of cadmium(Cd) shall be less than 0.01%.
-If the restricted substances concentration above the limits, the description, content of hazardous substances, and presence of component/part should be marked.

-中国 RoHS 限制物质的含量限值要求见 GB/T 26572 和 GB 26572，具体如下：铅(Pb)、汞(Hg)、六价铬(Cr^{6+})、多溴联苯(PBB)、多溴二苯醚(PBDE)、邻苯二甲酸二正丁酯(DBP)、邻苯二甲酸二异丁酯(DIBP)、邻苯二甲酸丁苄酯(BBP)、邻苯二甲酸二(2-乙基)己酯(DEHP)的含量应小于 0.1%，镉(Cd)的含量应小于 0.01%。

5.13.3 Labeling Requirements 标识要求

Parts or components used to produce final products in the scope of China RoHS are not mandated to implement labeling requirements. But supplier(s) shall provide hazardous substances information. Service spare parts are not mandated to implement separate labeling requirements.

用于生产中国 RoHS 范围最终产品的零部件或组件无需强制执行标识要求。但供应商应提供有害物质信息。服务备件无需实施单独的标识要求。

A hazardous substances table (see Table 2 below) shall be included in the Operator Manual for medical devices marked with logo 2 in “GD 21-A8 10433621-A8-AND-02S-xx: Specific Requirements for China”:
-The table shall be in Chinese language – any other language together with Chinese is acceptable.
-The presence or absence of China RoHS hazardous substances shall be stated at the highest component/part level.
-It shall be clearly specified to which system parts/components the hazardous substances table refers. The word “Other” should not be used in the table.

标有《GD 21-A8 10433621-A8-AND-02S-xx: 对中国的特定要求》中标志 2 的医疗设备的操作手册中应包括一个有害物质表（见下表 2）:

-该表应使用中文,中文和任何其他语言同时存在也可以接受。
-中国 RoHS 有害物质的存在或不存在应在最高的组件/部件级别上说明。
-应明确说明有害物质表涉及哪些系统部件/组件。表格中不得使用“其他”字样。

-每种中国 RoHS 有害物质的含量超过限值时，应以 "x "表示，不存在则以 "o "表示。

-When the presence exceeds concentration limit of hazardous substances shall be indicated by means of an “x”; the presence below concentration limit indicated by an “o”.
-When the presence of each hazardous substance in a component/part is below concentration limits, it is not required to be included in the table.
-Supplementary explanation may be added in Comment box if needed.
• Example 1: Explanation of technical reason for “x” in the Table, e.g. application of exceptions.
• Example 2: Additional information for optional extras.
-Supporting documents shall be maintained to prove information on the table. Supporting documents
shall be kept for a period of not less than three years after stopping production.
Other unmentioned refers to “Guidance on China RoHS requirements”& “SJ/T 11364 Marking for the restriction of the use of hazardous substances in electrical and electronic product ”.

-若某组件/零件中所有有害物质含量均低于限值，则无需列入表格。
-必要时可在备注栏添加补充说明。
• 示例 1：表格中“x”标记的技术原因说明（如使用了豁免）。
• 示例 2：可选配件的补充信息。

-证明表格信息的文档保存期应自产品停产后不少于三年。
其它未提及部件参见《中国 RoHS 要求指南》和《SJ/T 11364 电子电器产品有害物质限制使用标志要求》。

Table 2/表 2

产品中有害物质的名称及含量

部件名称 Parts	有害物质 Hazardous Substance									
	铅Pb	汞Hg	镉Cd	六价铬 Cr(VI)	多溴联苯 (PBBs	多溴二苯醚 PBDEs	邻苯二甲酸二正丁酯 DBP	邻苯二甲酸二异丁酯 DIBP	邻苯二甲酸丁苄酯 BBP	邻苯二甲酸二(2-乙基)己酯 DEHP
…….	…….	…….	…….	…….	…….	…….				

注1： O：表示该有害物质在该部件所有均质材料中的含量均不超出电器电子产品有害物质限制使用国家标准要求。
X：表示该有害物质至少在该部件的某一均质材料中的含量超出电器电子产品有害物质限制使用国家标准要求。
Note 1: O：Indicates that the content of the hazardous substance in all homogeneous materials of the part does not exceed the national standard for restricted use of hazardous substances in electrical and electronic products.
X：Indicates that the content of hazardous substance in at least one homogeneous material of the part exceeds the national standard for restricted use of hazardous substances in electrical and electronic products.
注2：以上未列出的部件，表明其有害物质含量均不超出电器电子产品有害物质限制使用国家标准要求。
Note 2: The parts not listed above indicate that the content of hazardous substances does not exceed the national standard for restricted use of hazardous substances in electrical and electronic products.

除非另外特别的标注,此标志为西门子医疗产品及其部件的环保使用期限。
此环保使用期限只适用于产品在产品手册中所规定的条件下工作。

The Environment-Friendly Use Period *(EFUP)* for Products of Siemens Healthineers and their parts is per the symbol shown here, unless otherwise marked. The EFUP is valid only when the product is operated under the conditions defined in the user documentation.

If a device contains parts with an EFUP shorter than the EFUP of the device itself (e.g. batteries), the parts
with shorter EFUP shall be marked separately and accompanying documents shall state that these parts have
to be exchanged within a time shorter than or equal to the respective part’s EFUP.
如果设备含有 EFUP 短于设备本身的 EFUP 的部件（如电池），EFUP 短的部件应单独标记，随附文件应说明这些部件必须在短于或等于各自部件 EFUP 的时间内更换。

5.14 Process validation / 过程确认

Transfer to production must be completed before process validation can begin. Series production samples, but no reworked drawing samples, are acceptable to use as evidence for a process validation. Process validation is described in 5521906-AND-812.	在流程确认开始前，生产转移必须先完成。 系列产品样例, 但不是重新翻新的图纸样机，可以用作过程确认的证据。 过程确认过程在 5521906-AND-812 中有详细描述。

5.15 Risk management / 风险管理

The product risk management (See 5521906-AND-807) is a process that is implemented over the entire life cycle of the product. The creation of the risk management plan for documenting and implementing the RM process as well as the risk assessment and documentation of the risk analysis and risk evaluation are to be started to define necessary risk mitigations to be implemented for further product development. If necessary, the risk analysis is updated, and the effectiveness of preventive measures reviewed in each process section.	产品风险管理是用于整个产品生命周期的一个过程（请参阅 5521906-AND-807）， 需要创建风险管理计划，以记录和实施风险管理（RM）过程，同时进行风险评估和风险分析的风险评估和记录，以定义必要的风险缓解措施，以便在产品开发的后续阶段实施。必要的时候，风险管理和预防措施的有效性需要在每个流程中进行更新。

5.16 Product Reliability/产品可靠性

The Product Reliability Management process shall minimize negative effect for users or customers due to limited reliability, availability, or maintainability of a product. The reliability related product requirements are handled through the development-V. For additional information see 11106101-AND-02S.	产品可靠性管理流程应尽量减少由于产品的可靠性、可用性或可维护性有限而对用户或客户的负面影响。 相关产品的可靠性要求通过开发-V 处理。 有关其他信息，请参阅 11106101-AND-02S。

5.17 Clinical evaluation / 临床评估

Clinical evaluation undertaken for the development of a medical device: Premarket research and development are guided by clinical evaluation and risk management. Typically, manufacturers carry out clinical evaluations to • define needs regarding clinical safety and clinical performance of the device; • in case of possible equivalence to an existing device, evaluate if there are clinical data	开展医疗器械的临床评估： 进入市场前的研究与开发要以临床评估和风险管理为指导。 通常，制造商用临床评估来： • 定义设备临床安全性和临床性能的需求； • 如果可能与现有设备相当，评估是否有临床数据可用并确定等效性； • 进行差距分析，确定被评估的设备仍然需要生成哪些数据，是否需要临床调查，如果是，定

available and determine equivalence; • carry out a gap analysis and define which data still need to be generated with the device under evaluation, whether clinical investigations are necessary and if so, to define the study design; As the initial clinical evaluation identifies the questions to be answered by a clinical investigation, the clinical evaluation process should generally commence in advance of any clinical investigation.	义研究设计； 由于最初的临床评估确定了临床调查要回答的问题，临床评估过程一般应在任何临床调查之前开始。

5.18 Project phase application/ 项目阶段申请

The project phase application documents the decision, if and how the next project phase may be started. Project phase applications submitted by the product line/project management, include • cross-references to the relevant documents forming the basis of the decision, • agreement from the head of the Business line that the relevant project (sub)section can be initiated.	项目阶段申请记录决定，是否及如何开展项目下一阶段。项目阶段申请由产品线/项目管理提出，包括： • 对用于支持做决定的相关文档的交叉参考 • 业务部门领导同意项目流程（子流程）可以启动

Project Application	Decision relating to	Applicant
M120	Start of product specification	产品经理
M150	Start of system specification	产品经理
M200	Start of subsystem development	项目经理
M300	Product release	项目经理
M400	Product phase-out	产品经理
M450 Review	EOS decision	产品经理
项目申请	关于	申请者
M120	产品规范的开始	产品经理
M150	系统规范的开始	产品经理
M200	子系统开发的开始	项目经理
M300	产品发布	项目经理
M400	产品退市	产品经理
M450 Review	停止产品支持的决定	产品经理

5.19 Handling Deviations/偏差处理

Deviations from the prescribed process have to be evaluated – as soon as discovered and measures have to be decided at least by product quality manager and project manager. It is also recommended to involve the process manager. They are documented e.g. in design review records, change requests or defects.	规定流程中如出现偏差需立刻进行评估，应对措施至少需由产品质量经理和项目经理决定。此外，还建议邀请流程经理。其记录在例如设计审查记录，变更请求或缺陷中。

6. Tasks and responsibilities / 任务和职责

6.1 GM (Head) of SSME MI / SSME MI 总经理

• ensures that the project-specific tasks, responsibility, and authority of the project manager are determined and documented	• 确保确定并写下项目相关工作、项目经理的职责和权限；
• makes M120 – M400 decisions regarding project requests, based on documents provided by the project manager	• 根据项目经理提供的文件，对项目申请作出决定(M120 – M400)；
• formally agrees to the request with his signature. Approval for the required resources (investments, personnel and financial resources) is granted with this signature	• 签名表示正式同意项目申请。批准项目所需的资源，并签名表示同意(投资、人事和财务资源)；
• Shall establish an independent quality management representative	• 确保建立一个独立的质量管理代表。

6.2 Quality Management Representative / 质量管理负责人

• confirms on project phase applications that the project plans are consistent with the quality management system currently in force.	• 对于项目阶段申请，确认质量计划与生效的质量管理体系的一致性；
• ensures that all standards and laws currently in force and applicable to the project are made available.	• 确保项目遵守所有生效的法律和法规；
• assigns an independent SSME CT Quality Control Engineer for each project/product	• 确保为每一个项目和产品有一个独立的质量控制工程师；

6.3 Project manager / 项目经理

• is responsible for design and development planning and tracking. • appoints the PSG members in consultation with their respective functional departments (definition and appointment in the PMP, QMP, or PQMP),	• 负责设计和开发计划和跟踪； • 与有关职能部门协商，任命项目管理小组成员（项目管理计划，质量管理计划，或 PQMP 的定义和任命）；
• Leads and motivates the product core team (members are project engineer, quality engineer, system engineer)	• 领导和激励产品的核心团队（成员包括项目工程师，质量工程师，系统工程师）；
• leads the PSG and acts as the representative for the overall project management,	• 带领项目管理小组，并且是整个项目的代表人；
• represents the interests of those functional departments not represented in the PSG,	• 承担代表未参加项目管理小组的那些职能部门利益的责任；
• ensures that, the PSG is made up of experts from the relevant functional departments	• 根据被牵涉的流程和任务，确保项目管理小组由各职能部门的专家组成；
• plans, manages and controls the project, including:	• 对项目进行计划、管理和控制；
• Preparing and updating the QMP/PMP as well	• 制定和更新质量管理计划和项目管理计划；

as document plans with support of document manager.	• 以及在文档经理协助下完成文档计划；
• Ensuring the project documentation is in place. • performing design reviews	• 确保项目文档的生成； • 执行设计评审；
• supporting patents application. • Ensuring that the necessary spare parts are planned in sufficient time. • Assuming responsibility for compliance with process requirements, for the creation and maintenance of the documentation in the DHF and DMR, and for the overall product quality achieved.	• 支持申请专利； • 确保有充足的时间计划准备必要的备件； • 承担符合流程要求，开发历史文档（DHF）和产品制造性文档（DMR）的创建和维护，整个产品质量达标的责任。

6.4 Quality Engineer / 质量工程师

• carries out checks on the project from an independent perspective and is an independent participant of design reviews. • Must stop processes and projects if the relevant quality requirements are not fulfilled (e.g. in terms of product features, intended use/intended purpose, or compliance with QM system) • The quality engineer escalates any relevant issues directly to the quality management representative of the SSME MI.	• 从独立的角度对项目进行检查，并且是设计评审的独立参与者； • 如果未满足相关质量要求（例如，在产品功能、预期用途/预期目的或符合 QM 体系方面），则必须停止流程和项目； • 质量工程师将任何相关问题直接上报给 SSME MI 的质量管理者代表。

6.5 Product steering group (PSG) / 项目管理小组成员

• Supports the project manager. • Remains responsible for the product even after the start of delivery, although the composition of the group can change. • Makes project decisions – one representative each affected functional area is essential. • Decides about any changes to milestones, changed work packages endangering the project timeline or phases, any incidents affecting the project or in case of occurrence of project risks • Decides whether milestones have to be repeated when changes are made to product scope. • Decides on any action to be taken as a result of the outcome from a design review	• 支持项目经理； • 尽管团队组成在产品交付后可能变化，仍对产品负责； • 做出项目决策 – 每个受影响的功能领域都必须有一名代表； • 决定里程碑的任何更改，危及项目时间表或阶段的工作包更改，影响项目的任何事件或发生项目风险的情况； • 决定在更改产品范围时是否必须重复里程碑； • 决定设计评审后采取的任何行动 。
Every member of the PSG • is a delegate representing their respective functional department, • coordinates tasks between the PSG and the respective functional department, and	每个项目管理小组的成员： • 是各个职能部门的代表； • 协调功能部门和 PSG 之间的任务，以及；

• makes project-relevant decisions for the respective functional department. • ensures that the regulations applicable to the project are implemented. • escalates topics to PSG which cannot be solved within the respective functional area, and upward along the hierarchical structures if it cannot be solved there. The tasks and responsibilities of the PSG members are specified in detail in the QMP.	• 为其所在的功能部门作与项目有关的决定； • 确保项目相关的规定得到实施； • 将不能在各功能区域内解决的问题汇报到 PSG，如仍未解决，汇报到更上层。 项目管理小组成员的任务和职责详细描述于质量管理计划（QMP）中。

7. Provisional solution and backward method / 过渡措施和补救办法

These instructions are applied to new projects on or after the effective date. For on-going projects before the design review R5, the QMP must be supplemented specifying how and when a transition will be made to the PEP. The transition must be completed 6 months after the validity date of these instructions at the latest. Process phases which have already been concluded through the design review do not need to be revised.	本规定适用于处于生效日期或在此之后的新项目。对于正在进行的、主评审 R5 之前的项目，必须在质量管理计划中补充说明如何以及何时进行 PEP 的过渡。最晚必须在本指令生效后的 6 个月内完成过渡。对于已经通过主评审而结束的过程阶段不必再进行修改。

8. Record / 记录

Record no. 记录编号	Record name 记录名称	Where stored. 保存地	Retention 保存期限
n.a.	n.a.	n.a.	n.a.

9. Appendix / 附录

Appendix no. 附录号	Appendix name 附录名称
1	R-Review Checklist/ 设计评审清单

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